

Amritanshu Kumar

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Professional Summary

Backend engineer specializing in **real-time distributed systems**. Owns and operates a live multiplayer platform serving **41K registered / ~3.4K daily active** users — owning the TypeScript socket server and its REST API end to end. Took the platform from single-instance to horizontally scaled: Redis-backed pub/sub fan-out, cross-instance leader election, distributed locking, and failover with zero session loss. **3,332 tests** under enforced coverage gates. Leads a 7-engineer team spanning client, server, and API.

Professional Experience

RootHoot Pvt Ltd

Team Lead | Backend Architecture

Remote, Hyderabad

Dec 2024 – Present

- Own the Node.js/TypeScript socket server and Express REST API for Pongz end to end — a live real-time multiplayer platform on Google Cloud, shipping on Android and iOS.
- Scaled the server horizontally from a single instance.** Redis pub/sub adapter for cross-instance event fan-out, per-instance identity, and room-routed delivery so any node can reach a socket held by any other.
- Solved matchmaking split-brain across N instances** with Redis leader election — the elected leader forms matches from a shared global queue instead of each node's local slice, with atomic per-player claims to make double-seating impossible under overlapping leases.
- Built multi-instance failover with zero session loss** — a draining node hands live games to its peers, which adopt them mid-session rather than dropping players. Validated by a real-Redis two-instance E2E suite.
- Engineered a **server-authoritative protocol** that eliminated race conditions from latency and message reordering — monotonic sequencing, fingerprint-based dedup on reconnect, and exponential backoff with jitter.
- Built **live-ops infrastructure** — graceful maintenance windows with coordinated shutdown, player draining, and admission gating so deploys land without killing in-flight sessions — plus the **competitive ranking and skill-based matchmaking** systems end to end (rating points, tier progression, pool formation with rank-step widening under low concurrency).
- Grew the suite to **3,332 passing tests** with CI-enforced coverage floors (80% lines / 85% functions / 70% branches), including deterministic real-socket and multi-instance E2E harnesses.
- Directed architecture, code review, and sprint planning across a **7-engineer team** spanning client, server, and API.

DeftSoft

Software Engineer

Mohali, Punjab

Feb 2023 – Oct 2024

- Shipped a product to **5M+ downloads / 4.6 stars** on Google Play; post-launch telemetry iteration lifted engagement **30%**.
- Introduced **TDD** across the team to catch regressions before release — ratings improved **1.5 stars** as a direct result. Authored a **25-chapter** technical course on software architecture.
- Profiled and optimized hot paths for **25%** higher throughput on low-end hardware. Mentored 3 junior engineers — code review, pair programming, and coding standards.

Selected Projects

Nimirta Engine: 2D physics engine from first principles, C++/SFML — impulse solver with warm starting, Baumgarte stabilization for high-velocity contacts, semi-implicit Euler integration, fixed timestep with spiral-of-death prevention, 3 collision algorithms, hand-written vector math.

InstaReload: Hot-reload toolchain cutting reload latency from 3–5s to **7ms**. Runtime **IL patching** via Roslyn for the common path (90% of edits), MonoMod detours for structural changes.

FlowUI: Open-source framework (MIT) eliminating UI boilerplate — CRTP builders, **Roslyn source generation** with partial classes, bidirectional O(1) lookups, 8-tab visual editor. Fully documented.

Technical Skills

- Languages:** TypeScript, C#, C++, Go, Python
- Backend:** Node.js, Express, Socket.IO, MongoDB, Redis (ioredis), JWT, Winston
- Distributed Systems:** Horizontal scaling, Redis pub/sub, leader election, distributed locking, failover & draining, state synchronization, idempotency & dedup, split-brain resolution
- Cloud & Ops:** Google Cloud (App Engine), Docker, GitHub Actions, structured logging, live-ops
- Testing:** Mocha, Chai, Sinon, c8 coverage gates, deterministic E2E harnesses, TDD

Education & Certifications

Bachelor of Computer Application, Lovely Professional University, Punjab, India | Sep 2020 – Jun 2023

Certifications: Unity Certified Programmer | Unreal Engine Game Design & Development Specialization